

Week 1 Status Report: Data Understanding & EDA Kickoff

Project Title: Daikin Air Conditioning Demand Forecasting & Supply Chain Optimization

Date: August 23

Report Owner: Sai Prashanth

Team Members & Roles: Arvind, Hariharan, Vani, Vedika, Sai Prashanth

Dataset Analyzed: Daikin_Filtered_2022 July to 2024 Mar.xlsx (82,033 clean transaction records spanning July 16, 2022 to June 30, 2024)

1. Executive Summary & Responsibility Allocation

This report consolidates the Week 1 findings for the Daikin Demand Forecasting project. The objective of this phase is to establish a rigorous data foundation, evaluate target metrics, conduct exploratory data analysis (EDA), diagnose data quality anomalies, and formulate strategic queries for the Deloitte project sponsors.

Team Responsibility Matrix

Member	Assigned Responsibility	Primary Deliverable / Milestone
Sai Prashanth	Report Owner & Governance	Overall progress tracking, sponsor query aggregation, and executive report synthesis.
Hariharan	Demand & Target Analysis	Evaluation of order/demand dynamics, IDU vs. ODU splits, and target variable definition.
Vani	Data Quality & Integrity Audit	Missing-value analysis, schema anomaly detection, and sales return quantification.
Vedika	SKU-Level EDA & Intermittency	Syntetos-Boylan demand classification, distributional

		profiling, and top-volume SKU analysis.
Arvind	Architecture & Pipeline Specs	Data dictionary creation, data type mapping, and ETL pipeline normalization rules.

2. Key Data Metrics Summary

Metric	Value	Technical Context
Total Transaction Records	82,033	Clean billing records (1 null row dropped).
Date Horizon	July 16, 2022 – June 30, 2024	625 unique transaction calendar days across 3 fiscal years.
Net Total Volume	902,391.0 units	Gross shipments (909,451.5) minus sales returns (−7,060.5).
Active SKU Portfolio	154 SKUs	Room Air Conditioner (RA) material codes.
Sales Offices (Depots)	5 Hubs	MAA (Chennai), SBD, SBD1, COK (Cochin), BLR (Bengaluru).
Lumpy Demand Share	70.1% of SKUs	108 out of 154 SKUs exhibit intermittent and erratic demand.

3. Data Dictionary & Schema Documentation (Arvind)

The transactional dataset comprises 23 raw fields, of which 4 columns represent redundant

system duplicates and ² fields are static enterprise constants.

Column Name	Storage Type	Null Count (%)	Cardinality	Business Definition & Feature Engineering Rule
Material	Categorical (str)	0 (0.0%)	154	Unique material identification code. Primary SKU key.
Billing Doc.	Numeric (float64)	23,603 (28.77	39,550	SAP ERP Invoice Number. Null values correspond to Q1 FY24-25.
Item	Numeric (float64)	23,603 (28.77	73	Line-item sequence number within the billing invoice.
SALES OFFICE CODE	Categorical (str)	0 (0.0%)	5	Regional sales office hub identifier (MAA, SBD, SBD1, COK, BLR).
EXT MAT GROUP	Categorical (str)	0 (0.0%)	2	Sub-assembly category: RA-ODU (Outdoor Unit) vs. RA-IDU (Indoor Unit).
Segment	Categorical	0 (0.0%)	2	Technology

	(str)			classification: Inverter vs. Non Inv.
Star Rating	Categorical (str)	0 (0.0%)	6	SAP internal rating code (I3S, I5S, N3S, N1S, N2S, IOT).
Star Rating.1	Categorical (str)	0 (0.0%)	5	Commercial star rating label (3 Star, 5 Star, 1 Star, 2 Star, 3 star).
Stor. Location	Numeric (float64)	23, 603 (28.77	1	Warehouse storage location code (1030.0).
Billing Date	Temporal (datetime64)	0 (0.0%)	625	Transaction posting date (daily level).
Customer	Categorical (str)	0 (0.0%)	2, 352	Account key for distributor or commercial purchasing entity.
Dealer Name	Categorical (str)	23, 603 (28.77	510	Retail dealer entity fulfilling end-user orders.
Tonnage	Numeric (float64)	0 (0.0%)	9	Cooling capacity (0.75, 1.0, 1.5, 1).
Billing Quantity ODU	Numeric (float64)	0 (0.0%)	189	Billed quantity of outdoor units.

Billing Quantity IDU	Numeric (float64)	0 (0.0%)	133	Billed quantity of indoor units.
Material.1	Categorical (str)	0 (0.0%)	154	Redundant exact copy of Material. (To be dropped)
SALES OFFICE CODE.1	Categorical (str)	0 (0.0%)	5	Redundant exact copy of SALES OFFICE CODE. (To be dropped)
Tonnage.1	Numeric (float64)	0 (0.0%)	9	Redundant exact copy of Tonnage. (To be dropped)
SBU / MatGroup3	Categorical (str)	0 (0.0%)	1	Invariant constant (RA). (To be dropped)

4. Data Quality & Missing-Value Audit (Vani)

4.1 Structural Temporal Missingness

An analysis of missing values reveals that missingness is not randomly distributed (MCAR), but rather exhibits a sharp structural boundary ($^{MAR} / ^{MNAR}$):

- Exactly **23,603 rows** (**28.77%** of the dataset) have missing entries for Billing Doc., Item, Stor. Location, and Dealer Name.
- Chronological Concentration:** **100%** of these null values occur strictly between **April 3, 2024 and June 30, 2024** (Q1 FY24-25).
- All transactions prior to April 3, 2024 contain **100%** completeness across invoice document metadata.

4.2 Sales Returns and Negative Adjustments

- **Negative Quantity Rows:** 1,735 transactions record negative quantities, representing sales returns, order cancellations, or SAP financial adjustments.
- **Volume Impact:** Total negative volume equals $-7,060.5$ units, against a gross positive volume of $909,451.5$ units (0.78% return rate).
- **Distribution:** Returns are primarily concentrated in top-volume inverter models (I3S and I5S star ratings) across MAA, BLR, and SBD sales offices.

4.3 Redundancies and Zero Volumes

- 243 rows register 0.0 units for both Billing Quantity ODU and Billing Quantity IDU.
- Star Rating contains minor text formatting inconsistencies (e.g., 3 Star vs. 3 star and SAP internal codes I3S, N3S, IOT).

5. Demand Dynamics & Target Variable Formulation (Hariharan)

5.1 Component Asymmetry (ODU vs. IDU)

The dataset reflects a pronounced structural imbalance between component dispatches:

- **Outdoor Units (RA-ODU):** $729,297.0$ net units (80.8% of overall volume) across 62,605 line items.
- **Indoor Units (RA-IDU):** $173,094.0$ net units (19.2% of overall volume) across 19,428 line items.

This indicates that split air conditioner systems are billed and dispatched as unbundled component sub-assemblies in SAP rather than unified 1:1 matched pairs.

5.2 Regional Volume Distribution

Sales Office Code	Location / Region	Total Net Units	Volume Share (%)	Mean Batch Size (Units)
MAA	Chennai Central Hub	323,256.0	35.82%	11.21
SBD	South Depot / Branch	244,248.5	27.07%	17.48

SBD1	South Depot Sub-Branch	151,400.0	16.78%	11.02
COK	Cochin Hub	99,905.0	11.07%	6.61
BLR	Bengaluru Hub	83,581.5	9.26%	8.05

5.3 Target Variable Definition

To support supply chain planning and safety stock allocation, the primary target variable $y_{i,j,t}$ is defined as:

$$y_{i,j,t} = \text{Billing Quantity ODU}_{i,j,t} + \text{Billing Quantity IDU}_{i,j,t}$$

where i represents the SKU (Material), j represents the regional distribution hub (SALES OFFICE CODE), and t represents the weekly temporal aggregation window (Week).

6. SKU-Level EDA & Intermittency Profiling (Vedika)

Demand intermittency across all 154 SKUs was evaluated using the **Syntetos-Boylan Classification** matrix. Metrics were calculated across daily time buckets:

1. **Average Demand Interval (ADI):**

$$ADI = \frac{N}{\sum_{t=1}^N \mathbb{I}(D_t > 0)}$$

2. **Squared Coefficient of Variation (CV^2):**

$$CV^2 = \left(\frac{\sigma_{D|D>0}}{\mu_{D|D>0}} \right)^2$$

Syntetos-Boylan Classification Cut-offs

Intermittency Cut-off: $ADI = 1.32$, Variability Cut-off: $CV^2 = 0.49$

CV ² < 0.49		CV ² ≥ 0.49	
+-----+-----+			
ADI < 1.32	SMOOTH	ERRATIC	
	(0 SKUs, 0%)	(17 SKUs, 11%)	
+-----+-----+			
ADI ≥ 1.32	INTERMITTENT	LUMPY	
	(29 SKUs, 18.8%)	(108 SKUs, 70.1%)	
+-----+-----+			

Profile Breakdown & Modeling Strategies

- Lumpy Demand** (70.1% – 108 SKUs):
High intermittency ($ADI \geq 1.32$) and high volatility ($CV^2 \geq 0.49$). Standard linear models or standard ARIMA fail on these series. Strategy: Croston SBA (Syntetos-Boylan Approximation), Markov-state models, or Zero-Inflated Gradient Boosting (LightGBM/XGBoost).
- Intermittent Demand** (18.8% – 29 SKUs):
Sparse demand arrivals ($ADI \geq 1.32$) with steady order batch sizes ($CV^2 < 0.49$). Strategy: Standard Croston method or Wright's modification.
- Erratic Demand** (11.0% – 17 SKUs):
Frequent transactions ($ADI < 1.32$) but severe spikes in quantity ($CV^2 \geq 0.49$). Strategy: Deep learning architectures (Temporal Fusion Transformers) or quantile regression.
- Smooth Demand** (0.0% – 0 SKUs):
No single SKU qualifies as stationary, continuous demand at the daily resolution.

Top 10 High-Volume SKUs & Intermittency Metrics

Material Code	Net Volume (Units)	Syntetos Category	ADI	CV2	Technology	Segment / Star
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RKM50U V16VA	118,346.0	Erratic	1.20	1.50	1.5 TR	Inverter / 5 Star
RKL50UV 16VA	61,530.5	Lumpy	1.41	1.35	1.5 TR	Inverter / 3 Star
RKL50UV 16V3	59,334.5	Lumpy	1.54	3.06	1.5 TR	Inverter / 3 Star
RKL35UV 16W	50,553.0	Lumpy	1.36	1.87	1.0 TR	Inverter / 3 Star
RKM50U V16V	48,236.0	Lumpy	1.93	2.49	1.5 TR	Inverter / 5 Star
RKL50UV 16VAF	37,737.0	Erratic	1.10	0.72	1.5 TR	Inverter / 3 Star
RL50UV1 6U2	36,631.0	Lumpy	1.69	2.74	1.5 TR	Non Inv / 3 Star
FTKM50 UV16VA	30,832.0	Erratic	1.11	1.71	1.5 TR	Inverter / 5 Star
RKM35U V16W	26,349.5	Lumpy	1.36	1.65	1.0 TR	Inverter / 5 Star
RL35UV1 6W1	24,735.0	Lumpy	1.59	2.45	1.0 TR	Non Inv / 3 Star

7. Open Questions & Sponsor Clarifications for Deloitte (Sai)

To ensure alignment prior to model training in Phase 2, the following technical queries are formally submitted to Deloitte:

1. **Q1. Data Ingestion Shift for Q1 FY24-25 (April–June 2024):**

All 23,603 records between April 3, 2024 and June 30, 2024 lack Billing Doc., Item, Stor. Location, and Dealer Name. Does this quarterly block represent un-invoiced order pipeline data, secondary dealer sales off-take, or an export format transition from SAP ERP?

2. **Q2. Component Billing Asymmetry ($4 : 1$ ODU to IDU Ratio):**

Outdoor units (729,297 units) account for over $4\times$ the billed volume of indoor units (173,094 units). Are indoor units recorded under a separate billing entity, or should forecasting models treat IDU and ODU as independent demand streams?

3. **Q3. Sales Return Credit Processing:**

Should the 1,735 return records (-7,060.5 units) be netted against original historical invoice dates, or retained as negative offsets in the calendar week the credit posted?

4. **Q4. Model Granularity & Hierarchy Alignment:**

Will the production forecasting system deploy inventory replenishment targets at the

Weekly SKU $\times$ Regional Sales Office level, or at an aggregated National SBU level?

8. Next Steps & Week 2 Action Plan

1. **Data Preprocessing & Cleaning:** Implement automated ETL script to prune duplicate columns (Material.1, SALES OFFICE CODE.1, Tonnage.1), handle zero-quantity rows, and standardize star rating strings.
2. **Feature Engineering:** Construct temporal features (lag variables, rolling seasonal means, cooling degree day proxies) and hierarchical aggregate keys.
3. **Baseline Modeling:** Benchmark intermittent demand baseline models (Croston SBA, Naive, Moving Average) against LightGBM regressors at the weekly level.